

TreasureDAO-9 Preliminary Audit Report (external, private)

⛑️ TreasureDAO-9 Fixes

Initial Code Freeze Commit Hash: Commit bf00640

Context

Issues Found

Severity	Tag	Count
Critical	(C-1)	-
High	(H-1)	1
Medium	(M-1)	1
Low	(L-1)	1
Code Quality	(Q-1)	-
Gas Optimization	(G-1)	-
Informational	(I-1)	-

[H-1] Seller and protocol owner may prevent buyer from exercising the option

Topic	Spec
Impact	High
Likelihood	High

In the `NFTCallOption` contract, `exerciseOption()` implementation contains important state changes, transfer of underlying NFT token to the buyer, and distribution of the proportional parts of strike price to the option writer and protocol owner.

```
function exerciseOption(uint256 optionId) external payable
nonReentrant { ... // Send protocol fee to owner if (protocolFee >
0) { (bool feeSuccess, ) = owner().call{value: protocolFee}("");
require(feeSuccess, "Protocol fee transfer failed"); } // Send
strike price (minus fee) to option writer (bool success, ) =
option.writer.call{value: writerPayment}(""); require(success,
"Strike payment transfer failed"); ... }
```

However, both protocol owner and more importantly option writer have capability of preventing option buyer from exercising the option by intentionally reverting on `call()` invocation. As a result, option writer can receive option premium and still be able to reclaim his NFT in any situation after option expires.

Remediations to consider

- Use pull instead of push approach for distributing assets. In this case both option writer and protocol owner should claim their corresponding proceeds from option purchase and exercise directly from the contract, instead of those assets being pushed to them.

[M-1] Premium and strike protocol fees can change

Topic	Spec
Impact	Medium
Likelihood	Medium

In the `NFTCallOption` contract, option buyer has capability to inspect current premium and strike protocol fees which are capped to max 5% of the premium and strike values correspondingly, but set to 2% and 0.25% respectively at the moment. In concrete case, option writer buy creating an option implicitly agrees to current fees. In addition, option writer has ability in case fees change and option has not yet been purchased to cancel the option.

However, when option is purchased option writer does not have any means to reject fee updates (especially strike protocol fee) that are disadvantageous to him.

Remediations to consider

- Record protocol fees as part of the option record, and respect these fees when they are applied in case they are more beneficial to the option writer than the currently configured protocol fees on the contract level.

[L-1] isExpired() may return misleading results

Topic	Spec
Impact	Low
Likelihood	Low

In the `NFTCallOption` contract, `isExpired()` is implemented in the following way:

```
/** * @notice Check if an option is expired * @param optionId ID of
the option */ function isExpired(uint256 optionId) external view
```